

Permutations & Combinations

COMPLETE LEARNING GUIDE

What You'll Learn

Permutations and combinations questions test one core judgment call above all: does the order of selection matter? Arrangements (permutations) count differently ordered outcomes as distinct, while selections (combinations) treat them as identical, and misidentifying which one a word problem needs is the single most common error on this topic. Beyond that basic distinction, placement tests layer in repeated objects (like letters in a word), circular seating, grouping constraints ('always together' or 'never together'), and 'at least one' conditions best solved through complementary counting rather than direct enumeration. This sub-module builds from the fundamental counting principle through permutations, combinations, circular arrangements, restricted arrangements, word-letter problems, and complementary counting — each with a fully worked numerical example — so that recognizing the right technique becomes a fast, reliable first step rather than a source of hesitation.

1. The Fundamental Counting Principle

If one choice can be made in m ways and a second, independent choice can be made in n ways, the two together can be made in $m \times n$ ways — this multiplication principle extends to any number of sequential independent choices and underlies every permutation and combination formula.

- A restaurant offers 4 starters, 5 main courses, and 3 desserts. The number of distinct 3-course meals possible = $4 \times 5 \times 3 = 60$.

2. Permutations: Arranging Distinct Objects (nPr)

When order matters and all objects are distinct, the number of ways to arrange r objects chosen from n is $nPr = \frac{n!}{(n-r)!}$. This is used whenever a question asks for the number of distinct SEQUENCES or ORDERINGS — such as ranking, seating in a row with labeled seats, or assigning distinct roles.

- In how many ways can 3 of 6 distinct books be arranged on a shelf? $6P3 = \frac{6!}{(6-3)!} = 6 \times 5 \times 4 = 120$.

3. Permutations With Repeated (Identical) Objects

When some objects being arranged are identical, the total arrangements $n!$ must be divided by the factorial of each repeated group's count, since swapping identical items doesn't create a genuinely new arrangement: $\text{total} = n!/(p! \times q! \times \dots)$.

- Find the number of distinct arrangements of the letters in the word BANANA (6 letters: B×1, A×3, N×2). Arrangements = $6!/(3! \times 2!) = 720/(6 \times 2) = 720/12 = 60$.

4. Circular Permutations

Arranging n distinct objects around a circle gives $(n - 1)!$ distinct arrangements, not $n!$ — because rotating the entire circle produces the same relative arrangement, effectively fixing one object's position as a reference. If clockwise and counterclockwise arrangements are considered identical (as with a necklace), divide by an additional factor of 2.

- 5 people are seated around a round table. Distinct seating arrangements = $(5 - 1)! = 4! = 24$. If the table were actually a symmetric necklace where flipping it over doesn't create a new arrangement, this would become $24/2 = 12$.

5. Combinations: Selecting Without Order (nCr)

When order does NOT matter — selecting a committee, a team, or any unordered group — use $nCr = n!/(r!(n - r)!) = nPr/r!$. The division by $r!$ removes the overcounting of the $r!$ different orderings of the same selected group.

- In how many ways can a committee of 3 students be chosen from 8? $8C3 = 8!/(3! \times 5!) = (8 \times 7 \times 6)/(3 \times 2 \times 1) = 336/6 = 56$.

6. Arrangements With Restrictions: Together, Never Together, and Alternating

For 'must be together' conditions, bundle the required group into a single unit, arrange the units, then multiply by the internal arrangements within the bundle. For 'no two of a certain type together' conditions, arrange the unrestricted objects first to create gaps, then place the restricted objects into those gaps.

- 5 men and 3 women are to be seated in a row of 8 chairs so that no two women sit together. Arrange the 5 men first: $5! = 120$ ways, creating 6 gaps (including the ends). Choose and arrange 3 of those 6 gaps for the women: $6P3 = 120$. Total arrangements = $120 \times 120 = 14400$.

7. Word-Letter Arrangement Problems: Vowels Together

When a question requires certain letters (often vowels) to always appear together within a word arrangement, treat the group of vowels as a single block, arrange it alongside the remaining letters (dividing by any repeated-letter factorials), and separately multiply by the internal arrangements of the vowels within their block.

- Find the number of arrangements of the letters in OFFICE (O,F,F,I,C,E — F repeats twice) such that all 3 vowels (O,I,E) are always together. Treat the vowel block as 1 unit alongside F,F,C — giving 4 units total, arranged in $4!/2! = 12$ ways (dividing by $2!$ for the repeated F). The 3 vowels within their block can be arranged in $3! = 6$ ways. Total = $12 \times 6 = 72$.

8. 'At Least One' Selections via Complementary Counting

For questions asking for selections with 'at least one' of a certain type, it is almost always faster to compute the total number of unrestricted selections and subtract the number of selections with NONE of that type, rather than directly summing every valid case (at least 1, at least 2, etc.).

- From a group of 5 men and 4 women, a committee of 4 is to be selected with at least 1 woman. Total ways (unrestricted) = $9C4 = 126$. Ways with NO woman = $5C4 = 5$. Ways with at least 1 woman = $126 - 5 = 121$.

Common Pitfalls

- Using the permutation formula (nPr) for a selection problem where order doesn't matter, or using combinations (nCr) when the question actually requires distinct orderings.
- Forgetting to divide by the factorial of each repeated letter's count when arranging a word with repeated letters.
- Using $n!$ instead of $(n-1)!$ for circular arrangements, and forgetting to divide by 2 when clockwise/counterclockwise arrangements are considered identical.
- In 'never together' problems, trying to directly count the valid arrangements instead of using total "together" cases, or making an error in the gap-method arrangement.
- Missing the complementary-counting shortcut for 'at least one' problems and instead attempting to enumerate every satisfying case directly, which is slower and more error-prone.
- When bundling a group as a single unit for a 'together' condition, forgetting to also multiply by the internal arrangements within that bundle.
- Confusing nCr with $nPr/r!$ conceptually — forgetting WHY the division by $r!$ happens (it removes the overcounted internal orderings of each selected group).

Key Takeaways

- Fundamental counting principle: sequential independent choices multiply — m ways $\times$ n ways = $m \times n$ total outcomes.
- $nPr = n!/(n-r)!$ for arrangements (order matters); $nCr = n!/(r!(n-r)!) = nPr/r!$ for selections (order doesn't matter).
- Circular arrangement of n distinct objects = $(n-1)!$; divide by 2 more if mirror-image arrangements count as identical.
- Word with repeated letters: total arrangements = $n! \div$ (product of each repeated letter's factorial).

- 'Together' conditions: bundle as one unit, arrange the units, then multiply by the bundle's internal arrangements.
- 'Never together' conditions: arrange the unrestricted items first to create gaps, then place restricted items into those gaps.
- 'At least one' is fastest via complement: total selections " selections with none of the required type.

Learn More

GeeksforGeeks — Permutation and Combination

<https://www.geeksforgeeks.org/maths/permutation-and-combination/>

IndiaBix — Permutation and Combination Aptitude Questions

<https://www.indiabix.com/aptitude/permutation-and-combination/>